

# Pulley Rescue Relay

*Lift the load with smarter force, not stronger arms*

**At a glance:** Lift the load with smarter force, not stronger arms. Plan about 60 minutes for the core block; 4 to 6 teams of 4 to 5 students.

## LEARNING OBJECTIVES

By the end of this activity, each student can:

- Explain the core science of this challenge: mechanical advantage, pulleys, force direction (Understand).
- Design a fair test by controlling one variable while holding others constant (Apply).
- Use their own data to decide whether a redesign improved the result (Analyze).
- Defend a final claim with specific evidence (Evaluate).

## MATERIALS FOR 25 STUDENTS AND 6 TEAMS

- Classroom pulleys
- String
- Weights
- Spring scales

## PREP BEFORE STUDENTS ARRIVE (15 TO 20 MIN)

- 1 Set up one station per team with the materials above, sorted and ready.
- 2 Stage the scoring sheet and any reference values before testing begins.
- 3 Put out PPE (gloves, goggles, trays) before any materials are handled.
- 4 Load the activity slides and ready the projected timer.

## MINUTE-BY-MINUTE FACILITATION

### Phase 1 Hook 0:00 - 0:05

Pose the mission as a challenge: "Lift the load with smarter force, not stronger arms." Take a quick prediction by show of hands.

### Phase 2 Prediction 0:05 - 0:12

Before any materials move, each team writes one prediction and one reason on the handout. No building yet.

### Phase 3 Constraints 0:12 - 0:18

Set the rules: approved materials only, change one variable at a time, record at least one data point before any redesign, and follow the safety notes.

### Phase 4 Build or solve 0:18 - 0:38

Teams work through the steps on the student handout. Circulate, ask what they are testing, and resist solving it for them.

### Phase 5 Test and record

Teams run the standard test and log a baseline result before changing anything.

### Phase 6 Redesign

Each team changes exactly one variable, predicts the effect, and re-tests to compare against baseline.

### Phase 7 Score, debrief, cleanup last 12 min

Teams submit a score slip, answer a debrief question with evidence, and reset the station. PPE off, wash or wipe down.

## TROUBLESHOOTING

**Problem:** A team changed several things at once. **Fix:** Have them reset to one variable before the next reading counts; treat it as a data-quality coaching moment.

**Problem:** No measurable result on the first test. **Fix:** Check the setup against the steps, confirm the standard test was followed, and log the baseline anyway.

**Problem:** Readings vary between trials. **Fix:** Standardize the procedure: same conditions, same technique, average several trials.

**Problem:** A team finishes early. **Fix:** Push the extension prompt below and ask them to quantify their improvement.

## DIFFERENTIATION: THREE-TIER SUPPORT PROMPTS

Tier 1 (stuck at the start):

- "What is the one science idea behind this challenge?" Point them to mechanical advantage.

Tier 2 (building but not reasoning):

- "What single change could improve your result without changing anything else?"

Tier 3 (ready to extend):

- "Run a second version that isolates a different variable and compare the two with data."

## DEBRIEF QUESTIONS (PICK 2 TO 3)

- Why did the movable pulley reduce effort?
- What did you trade for the easier lift?
- Where are pulley systems used in rescue work?

## SCORING RUBRIC (100 POINTS)

| Scoring criterion | Points     |
|-------------------|------------|
| Performance       | 35         |
| Setup accuracy    | 25         |
| Explanation       | 20         |
| Teamwork          | 20         |
| <b>Total</b>      | <b>100</b> |

## CLEANUP AND SOURCES

Return reusable items to the bin, dispose of consumables, wipe surfaces, and collect score slips.

**Sources:** TryEngineering Pulleys and Force